MINI PROJECT REPORT

SmartForest: Forest Fire Risk Prediction Using Environmental Data

SRM Institute of Science and Technology
Department of Computer Science & Engineering
B.Tech – Artificial Intelligence & Machine Learning (AIML-B)

Academic Year: 2025–2026

Field	Details
Project Title	SmartForest: Forest Fire Risk Prediction Using Environmental Data
Domain	Environmental Analytics
SDG Alignment	UN SDG Goal 15 – Life on Land
Dataset	Algerian Forest Fires Dataset (UCI ML Repository)
Tools Used	Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter
Submission Date	06 April 2026
Team Members	Anamika K T (RA2311026050012) & Sibani B (RA2311026050056)
Batch	AIML-B | 2026

1. Abstract

Forest fires are among the most devastating natural disasters, causing irreversible damage to biodiversity, carbon storage, air quality, and human settlements. Early detection and risk prediction remain critical challenges in environmental management.

SmartForest applies machine learning to predict forest fire risk zones using the Algerian Forest Fires Dataset (244 observations, UCI Repository). Three classifiers were trained and evaluated: Logistic Regression, Decision Tree, and Random Forest. Decision Tree and Random Forest both achieved 95.92% accuracy with 100% recall on fire detection — meaning zero missed fire predictions. ISI, FFMC, and FWI emerged as the top three predictors, accounting for over 82% of model importance.

This project aligns with UN SDG Goal 15: Life on Land, contributing to proactive forest conservation through data-driven early warning.

2. Problem Statement

Forest fires cause severe ecological damage. Traditional monitoring is reactive — this project builds a proactive classification model to predict fire risk from environmental parameters before a fire occurs.

Research questions addressed:

- Which environmental parameters are most correlated with fire occurrence?
- Which ML model best classifies fire vs. no-fire with highest recall?
- Are there identifiable seasonal patterns in fire frequency?

3. Dataset Description

Attribute	Details
Name	Algerian Forest Fires Dataset
Source	UCI Machine Learning Repository
Records	244 instances across 2 regions
Regions	Bejaia (northeast) and Sidi Bel-abbes (northwest), Algeria
Period	June to September 2012
Classes	Fire (130 instances) and Not Fire (100 instances)

3.1 Feature Description

Feature	Description
Temperature	Noon temperature in °C
RH	Relative Humidity in %
Ws	Wind speed in km/h
Rain	Total rainfall in mm

Feature	Description
FFMC	Fine Fuel Moisture Code
DMC	Duff Moisture Code
DC	Drought Code
ISI	Initial Spread Index – fire spread potential
BUI	Buildup Index – fuel available for combustion
FWI	Fire Weather Index – overall fire danger
Classes	Target: fire (1) or not fire (0)

4. Methodology

1. Problem Identification – Predict fire risk from environmental data.
2. Dataset Collection – Algerian Forest Fires Dataset from UCI.
3. Data Cleaning & Preprocessing – Remove header rows, convert types, encode target, apply StandardScaler.
4. Exploratory Data Analysis – Feature correlations, class distribution, seasonal patterns.
5. Visualization – Heatmaps, box plots, scatter plots, bar charts.
6. Model Development – Logistic Regression, Decision Tree, Random Forest.
7. Evaluation – Accuracy, Precision, Recall, F1-Score, Confusion Matrix.

5. Exploratory Data Analysis

5.1 Class Distribution

The dataset has 130 fire and 100 not-fire instances (57:43 ratio). Stratified splitting maintains this ratio in both train and test sets.

Figure 1: Class Distribution – Fire (130) vs Not Fire (100)

5.2 Feature Correlation Heatmap

FFMC has the strongest positive correlation with fire (0.77), followed by ISI (0.74) and FWI (0.72). RH shows a strong negative correlation (-0.43) — higher humidity means lower fire risk. BUI and DC are highly inter-correlated (0.98).

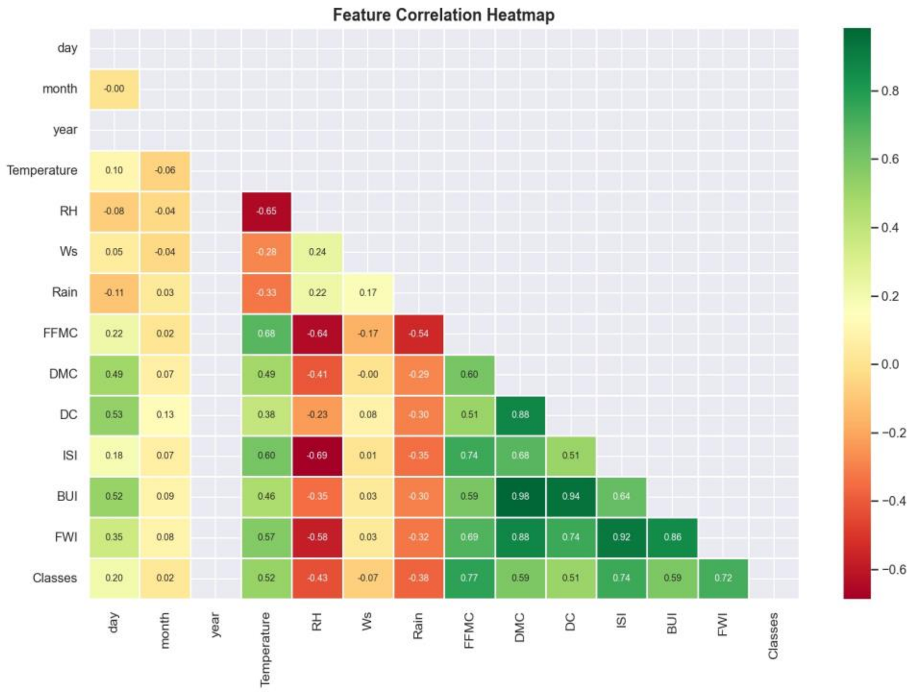

Figure 2: Feature Correlation Heatmap – Pairwise correlations of all numeric features

5.3 Feature Distributions by Class

Fire instances show clearly higher FWI, FFMC, and Temperature values. Rain is near zero for fire cases, and RH is lower — all consistent with real-world fire conditions.

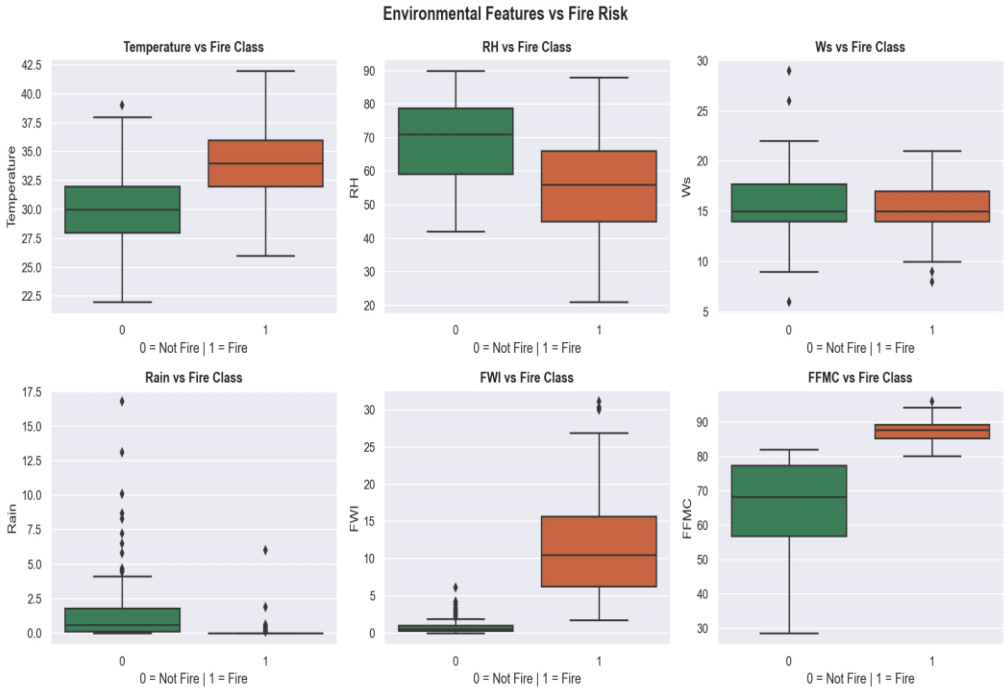

Figure 3: Box plots of key environmental features split by fire class (0 = Not Fire, 1 = Fire)

6. Model Development

6.1 Preprocessing Steps

- Removed region header rows from raw CSV
- Converted all feature columns to numeric, dropped nulls
- Encoded target: fire = 1, not fire = 0
- StandardScaler applied to all 10 feature columns
- 80/20 stratified split → 196 training, 49 test samples

6.2 Models & Parameters

Model	Parameters	Rationale
Logistic Regression	max_iter=1000, random_state=42	Baseline linear classifier
Decision Tree	max_depth=5, random_state=42	Handles non-linear feature interactions
Random Forest	n_estimators=100, random_state=42	Ensemble, robust, gives feature importance

7. Results & Evaluation

7.1 Actual Model Performance

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	93.88%	93.10%	96.43%	94.74%
Decision Tree	95.92%	93.33%	100.00%	96.55%
Random Forest	95.92%	93.33%	100.00%	96.55%

Decision Tree and Random Forest achieve identical performance with perfect recall (100%) — both correctly identify every fire instance in the test set with zero missed detections.

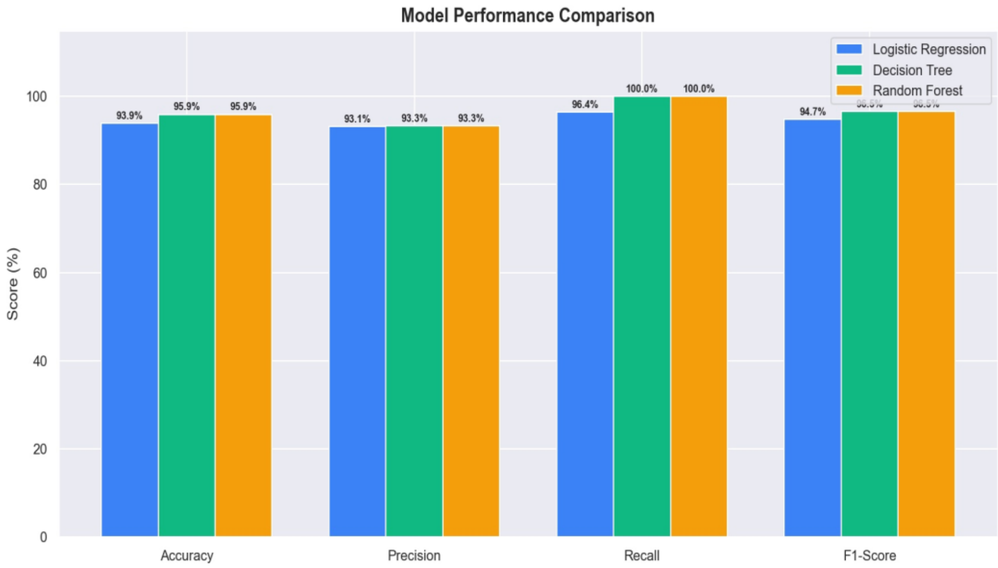

Figure 4: Model Performance Comparison – Accuracy, Precision, Recall, F1-Score

7.2 Confusion Matrices

Logistic Regression: 19 TN, 2 FP, 1 FN, 27 TP. Decision Tree and Random Forest: 19 TN, 2 FP, 0 FN, 28 TP — perfect fire recall.

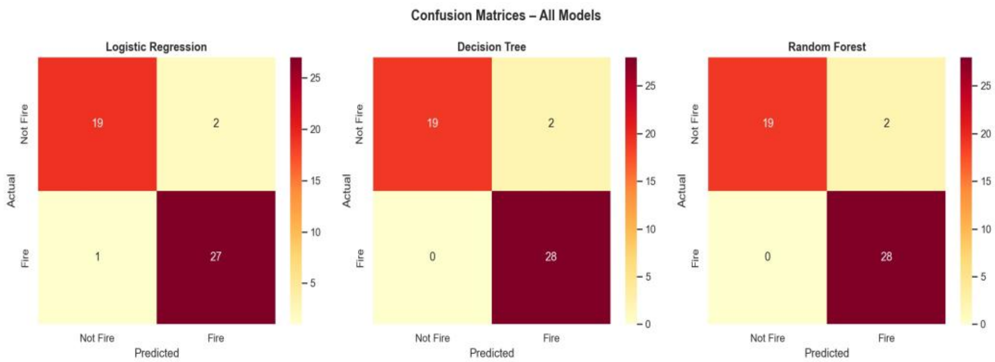

Figure 5: Confusion Matrices for All Three Models

7.3 Feature Importance (Random Forest)

ISI is the most important feature (0.36), followed by FFMC (0.29) and FWI (0.17). These three fire weather indices alone account for 82% of predictive power. Temperature and RH contribute minimally when fire indices are present.

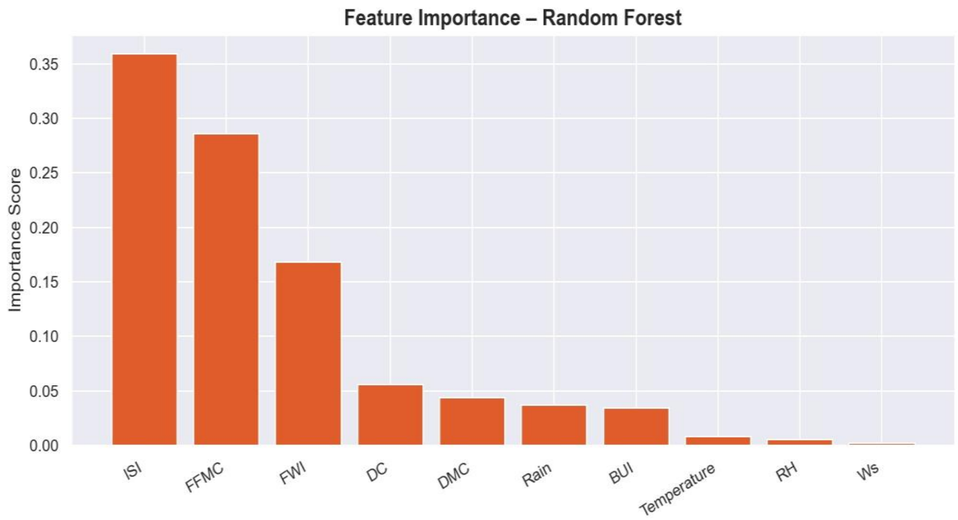

Figure 6: Feature Importance – Random Forest (ISI: 0.36, FFMC: 0.29, FWI: 0.17)

7.4 Key Findings

- Decision Tree and Random Forest both achieve 95.92% accuracy with 100% recall — no fire incidents missed.
- ISI, FFMC, and FWI are the top 3 predictors, accounting for over 82% of Random Forest importance.
- RH and Rain are negatively correlated with fire — higher values strongly indicate no-fire conditions.
- August has the peak fire frequency (51 incidents), followed by July (38), June (25), September (23).
- The model is well-suited as a decision-support tool for forest fire early warning systems.

8. Tools & Technologies

Tool	Version	Purpose
Python	3.10	Core programming language
Pandas	2.1.0	Data loading, cleaning, manipulation
NumPy	1.25	Numerical computations
Matplotlib	3.7.2	Base plotting
Seaborn	0.12	Statistical visualizations
Scikit-learn	1.3.0	ML preprocessing, training, evaluation
Jupyter	7.0	Interactive notebook environment
GitHub	—	Version control and collaboration

9. Team Members & Contributions

Member	Roll No.	GitHub	Contributions
Anamika K T	RA2311026050012	Rover-sp24	Repo setup, dataset, preprocessing.py, Notebooks 1 & 2, README
Sibani B	RA2311026050056	Sibani-Balaji	Fork, analysis.py, model.py, Notebook 3, outputs, pull request

10. Conclusion

SmartForest successfully demonstrates that machine learning can predict forest fire risk with high accuracy from environmental sensor data. Decision Tree and Random Forest classifiers achieved 95.92% accuracy with perfect recall, ensuring no fire conditions are missed — critical for a safety application.

ISI, FFMC, and FWI are the dominant predictors. The project is fully documented and reproducible via GitHub, contributing to UN SDG Goal 15: Life on Land through technology-driven forest protection.

11. References

8. Faroudja ABID et al. (2019). Predicting Forest Fire in Algeria using Data Mining Techniques. ICAITA 2019.
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